

SEPT 30, 2026

ALCF Facility API

BENOIT CÔTÉ

Software Developer

Argonne Leadership Computing Facility



Argonne National Laboratory
U.S. Department of Energy laboratory
managed by UChicago Argonne, LLC



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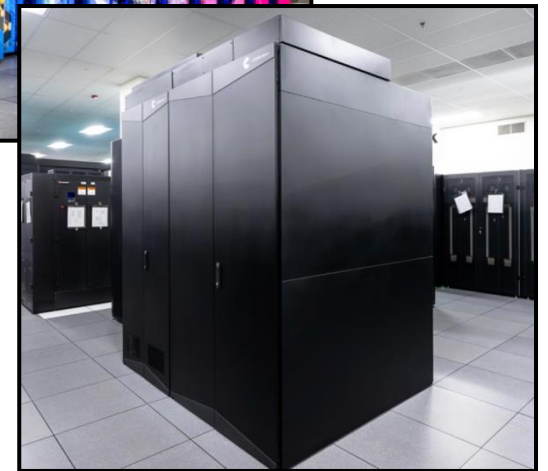


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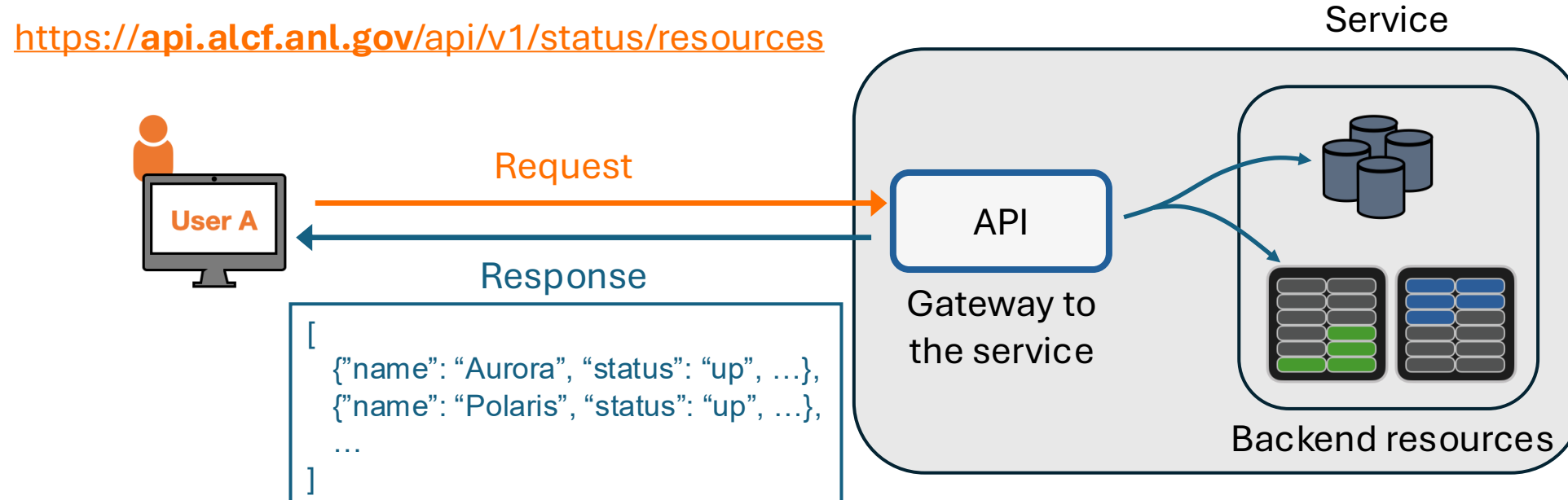
Session Plan

- Overview of the Facility API
 - Capabilities
 - Architecture
- Access to the Facility API
 - Authentication with Globus Auth
 - Exploration of alcf-tokens package
- Usage examples (exercises)
 - Discover resources
 - Submit and cancel jobs
 - Check job status and filter lists of jobs
 - Execute filesystem commands (view, ls, tail, head, chmod, ...)
 - List projects, query compute allocations, query storage allocations



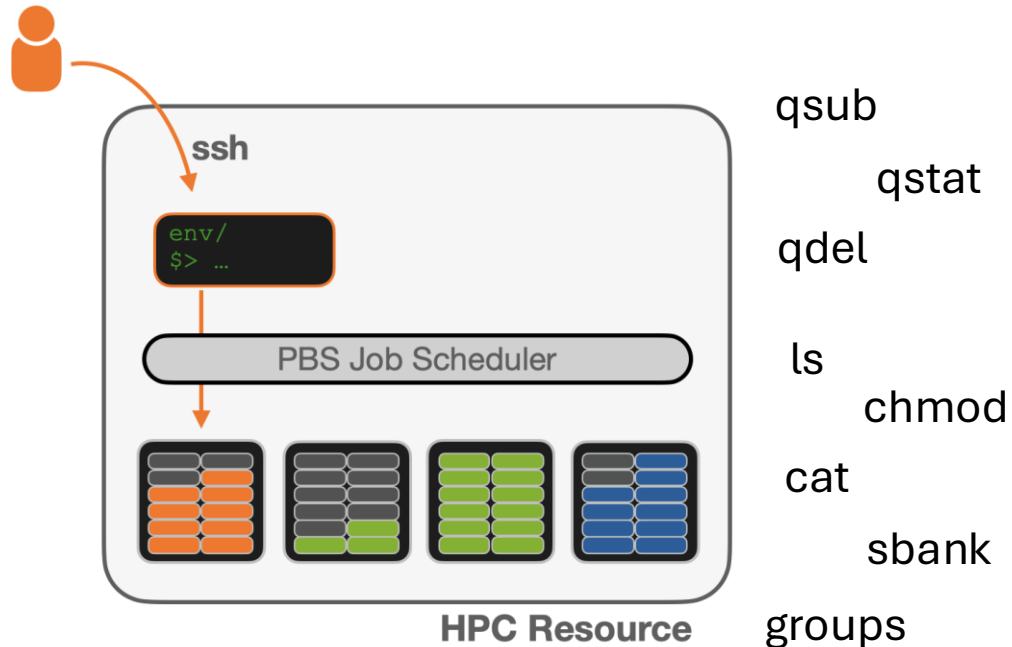
What is an API? (Application Programming Interface)

- An API allows users and applications to communicate with services to query information or to perform actions.
- It is done by sending an HTTP request to a specific URL, where the API is at the receiving end.
- An API typically returns structured data after it performed the action you requested.

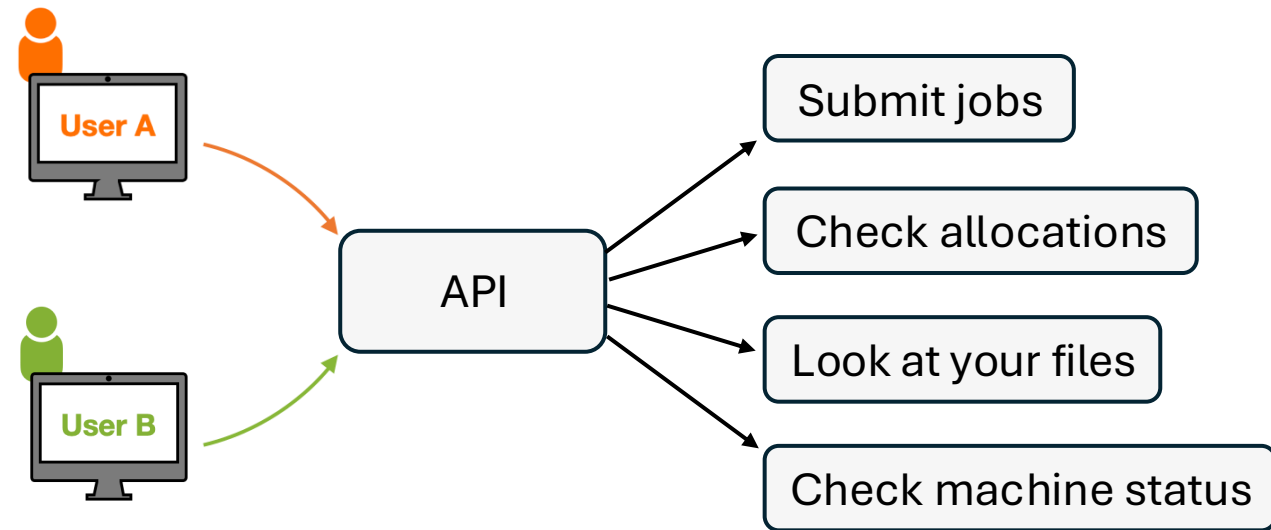


How Does the ALCF Facility API Work?

In the traditional approach, you would ssh into an HPC clusters to interact with it.



With the Facility API, once authenticated, you can access the ALCF resources without having to ssh into each HPC cluster.



Why is the API Useful?

- It allows to interface with ALCF resources **remotely and programmatically**.
- Requests to the ALCF Facility API can be embedded in scripts and workflows that interact with ALCF resources automatically. **Automated pipelines can therefore run without your intervention.**
- The ALCF Facility API can be made available to AI Agents to orchestrate workflows for you (see talks this afternoon).
- The Facility API is part of the Integrated Research Infrastructure (IRI) project, which aims to provide uniformed APIs with the same spec across DOE labs.

Facility API Components

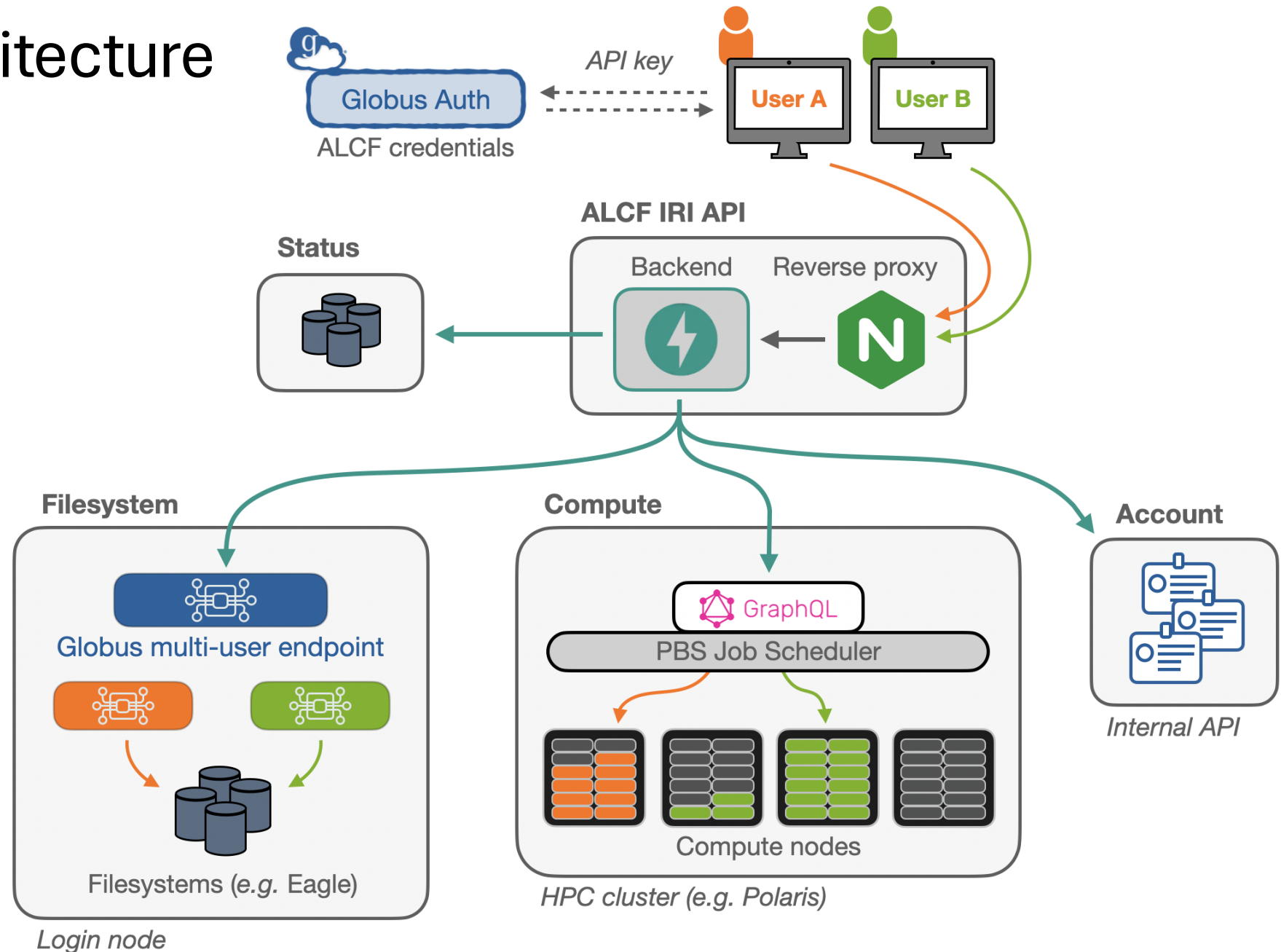
- **Facility**
 - Metadata on the ALCF Facility
- **Status**
 - Metadata and status of ALCF machines
- **Account**
 - Projects membership and allocations
- **Compute**
 - Submit and cancel jobs
 - Query state of jobs
- **Filesystem**
 - Submit asynchronous filesystem tasks
- **Task**
 - Query status of filesystem tasks
 - Get task results

facility		Limited set of examples	
GET	/api/v1/facility	Get Facility	✓
status			
GET	/api/v1/status/resources	Get all resources	✓
GET	/api/v1/status/resources/{resource_id}	Get a specific resource	✓
account			
GET	/api/v1/account/projects	Get the projects of the current user	✓
GET	/api/v1/account/projects/{project_id}/project_allocations	Get the allocations of the current user's projects	✓
compute			
POST	/api/v1/compute/job/{resource_id}	Submit Job	✓
GET	/api/v1/compute/status/{resource_id}/{job_id}	Get Job Status	✓
DELETE	/api/v1/compute/cancel/{resource_id}/{job_id}	Cancel Job	✓
filesystem			
PUT	/api/v1/filesystem/chmod/{resource_id}	Put Chmod	✓
GET	/api/v1/filesystem/ls/{resource_id}	Get Ls Async	✓
task			
GET	/api/v1/task/{task_id}	Get Task	✓

Swagger docs for complete list: <https://api.alcf.anl.gov>

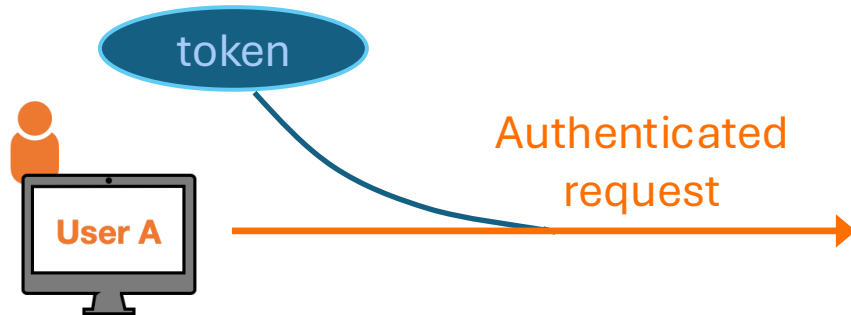
Facility API Architecture

- Compute requests are sent to GraphQL APIs that sit on top of our PBS schedulers.
- Filesystem commands are sent to a Globus multi-user endpoint that execute pre-canned functions.
- Account requests are re-routed to an internally-manage API.
- Status and metadata are maintained and stored in a local database.



Globus Auth

- Authentication and authorization platform (OAuth2/OpenID compliant)
- Federated identity provider integrating with different institutions worldwide
- From a user's perspective:
 - Auth generates **access tokens**
 - These tokens are sent to services to authenticate your requests



The screenshot shows the Globus Auth 'Log in' interface. The main screen prompts the user to 'Use your organizational login' with examples like 'university, national lab, facility, project'. A dropdown menu shows 'Argonne LCF' selected. Below this, it states 'By selecting Continue, you agree to Globus [terms of service](#) and [privacy policy](#).' A blue 'Continue' button is at the bottom. A blue arrow points from the 'Continue' button to a modal window on the right. The modal is titled 'Redirect to the selected identity provider's login page' and contains fields for 'ALCF Username', 'Password', and 'Cryptocard or Mobile token password'. A blue 'SIGN IN' button is at the bottom of the modal. To the right of the main login screen, text indicates '1100+ identity providers' and 'The Facility API requires ALCF credentials'.

Authenticate using alcf-tokens

https://github.com/argonne-lcf/Service_Enabled_Science

```
./setup.sh
```

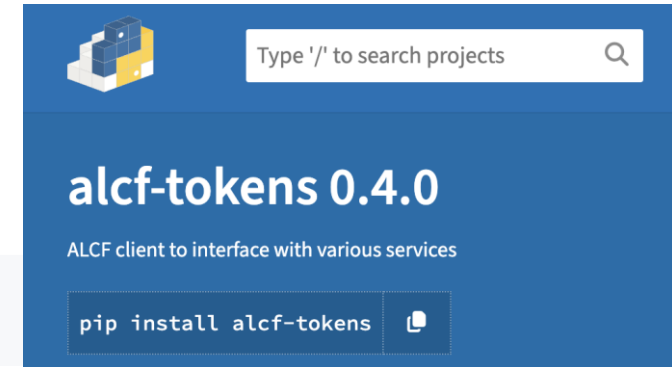
Activate your virtual environment:

```
source .venv/bin/activate
```

To properly gain access to all ALCF services, **make sure you authenticate with your ALCF credentials.**

```
alcf-tokens login --authorize-transfer home --authorize-transfer eagle
```

- This will generate a URL that you can copy-paste in your internet browser.
- Go through the Globus Auth login flow (**with your ALCF credentials**) and copy-paste the resulting authorization code back to your terminal.
- This will generate all the tokens needed for the rest of this hackathon!



Use Your Facility (IRI) API Token

Shell

```
access_token=$(alcf-tokens get-token iri)

curl -X POST "https://api.alcf.anl.gov/api/v1/compute/job/${resource_id}" \
  -H "Authorization: Bearer ${access_token}" \
  -H "Content-Type: application/json" \
```

Python

```
import requests
from alcf_tokens.auth import get_access_token

access_token = get_access_token("iri")
headers = {
    "Authorization": f"Bearer {access_token}",
    "Content-Type": "application/json"
}

response = requests.post(
    f"https://api.alcf.anl.gov/api/v1/compute/status/{resource_id}",
    headers=headers
)
```

List of Exercises on GitHub (see README file)

git clone https://github.com/argonne-lcf/Service_Enabled_Science.git

- 01_get_resources.py (*no auth required*)
- 02_get_resource.py (*no auth required*)
- 03_submit_job.py
- 04_get_job_state.py
- 05_view_file.py
- 06_list_jobs.py
- 07_cancel_job.py
- 08_get_projects.py
- 09_get_project.py
- 10_get_allocations.py

Supported filesystems

Eagle, Polaris:Home

Supported HPC cluster

Polaris, Crux, (and Aurora, not used today)

Compute allocation

alcf_training

Reservations

Polaris: R7645913

Crux: R314927

Edit script

Add your ALCF username to 03_submit_job.py

Documentation and Resources

- Hackathon materials
 - https://github.com/argonne-lcf/Service_Enabled_Science/
- ALCF official usage documentation
 - <https://docs.alcf.anl.gov/services/iri-api/>
- Swagger API documentation
 - <https://api.alcf.anl.gov/>
- OpenAPI spec
 - <https://api.alcf.anl.gov/openapi.json>
- Getting help
 - Email support@alcf.anl.gov